MINI-PROJECT • TRANSFER LEARNING REPORT

Transfer Learning for Image Classification

A First-Principles Comparative Study on ResNet50 vs. DenseNet121 Architectures & Fine-Tuning Dynamics

AUTHOR	FRAMEWORK	DATASET	TOP ACCURACY
Govardhan	PyTorch 2.x & Torchvision	Pizza, Steak, Sushi (3 Classes)	96.0% (ResNet50 Fine-Tuned)

1. First-Principles Mechanics of Transfer Learning

Deep Convolutional Neural Networks (CNNs) trained on vast image corpora like ImageNet (1.2M images) construct a hierarchical feature extraction funnel. From first principles, early convolutional layers learn universal spatial primitives—such as Gabor-like directional edges, color gradients, and pixel contrast transitions. Intermediate layers combine these primitives into geometric textures and motifs, while deep layers abstract these motifs into semantic object components.

Transfer learning leverages this hierarchy by reusing frozen low-level feature extractors for target domain tasks with limited training samples (225 images across Pizza, Steak, and Sushi). Reusing pre-trained weights dramatically reduces model variance, eliminates the need to train millions of weights from scratch, and prevents severe overfitting.

2. Structural Differences: ResNet50 vs. DenseNet121

ResNet50: Additive Residual Learning

Resolves vanishing gradients in deep networks by introducing **shortcut residual connections**. Rather than learning an unguided transformation function H(x), stacked layers learn a residual mapping F(x).

$$\text{Output} = F(x) + x$$

The identity addition (+ x) creates an uninterrupted gradient pathway back to early layers during fine-tuning.

DenseNet121: Feature Map Concatenation

Connects every layer to all subsequent layers inside a dense block. Information flows through direct feature map concatenation rather than element-wise addition.

$$\text{Output} = \text{Layer}_l\left([x_0, x_1, x_2, \dots x_{l-1}] \right)$$

Maximizes feature reuse across layers and reduces parameter count, though channel concatenation increases memory footprint.

3. Data Preprocessing & Augmentation Pipeline

Input images are resized to 224×224 pixels and normalized using ImageNet channel statistics (Mean: [0.485, 0.456, 0.406], Std: [0.229, 0.224, 0.225]). To broaden feature coverage on our small training set, data augmentations—including random horizontal flipping, ±15° rotation, and color jitter—were applied to enforce invariance against scale, orientation, and lighting variations.

4. Training Strategy & Fine-Tuning Execution

Training was executed across two distinct phases to maintain training stability:

- **Phase 1 (Feature Extraction):** Base convolutional layers were frozen. Only the new classification heads (Linear → ReLU → Dropout(0.3) → Linear) were trained using Adam (Learning Rate = 0.001) for 5 epochs.
- **Phase 2 (Selective Fine-Tuning):** Deep feature blocks—specifically `layer4` for ResNet50 and `denseblock4` for DenseNet121—were unfrozen. A lower learning rate (Learning Rate = 0.0001) was applied to adapt deep feature maps to food-specific textures without disrupting earlier pre-trained layers.

5. Model Evaluation & Performance Comparison

Both models were evaluated on an independent test set of 75 images (25 Pizza, 19 Steak, 31 Sushi). The detailed classification metrics are summarized below:

MODEL ARCHITECTURE	CLASS	PRECISION	RECALL	F1-SCORE	OVERALL ACCURACY
ResNet50 (Fine-Tuned `layer4`)	Pizza	1.00	0.96	0.98	**96.0%**
	Steak	0.90	0.95	0.92	
	Sushi	0.97	0.97	0.97	
DenseNet121 (Fine-Tuned `denseblock4`)	Pizza	1.00	0.84	0.91	**91.0%**
	Steak	0.76	1.00	0.86	
	Sushi	0.97	0.90	0.93	

6. Confusion Matrix Visual Analysis

The side-by-side confusion matrices highlight prediction distribution and class misclassifications across both architectures:



7. Key Findings & Architectural Insights

ResNet50 achieved superior accuracy (96% vs. 91%) and macro F1-score (0.96 vs. 0.90) compared to DenseNet121. ResNet's additive residual connections allow gradient updates to propagate cleanly through identity shortcuts, maintaining feature stability during fine-tuning. In contrast, DenseNet's channel concatenation introduces high-dimensional feature accumulation, making fine-tuning on smaller datasets slightly more sensitive to noise.